

Milk consumption: aggravating factor of acne and promoter of chronic diseases of Western societies.

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Consumption of cow's milk and cow's milk protein result in changes of the hormonal axis of insulin, growth hormone and insulin-like growth factor-1(IGF-1) in humans. Milk consumption raises IGF-1 serum levels in the perinatal period, adolescence and adulthood. During puberty with the physiological onset of increased secretion of growth hormone, IGF-1 serum levels increase and are further enhanced by milk consumption. IGF-1 is a potent mitogen; after binding to its receptor in various tissues, it induces cell proliferation and inhibits apoptosis. Keratinocytes and sebocytes, as well as the androgen-synthesizing adrenals and gonads, are stimulated by IGF-1. The epidemic incidence of adolescent acne in Western milk-consuming societies can be explained by the increased insulin- and IGF-1-stimulation of sebaceous glands mediated by milk consumption. Acne can be regarded as a model for chronic Western diseases with pathologically increased IGF-1-stimulation. Many other organs, such as the thymus, bones, all glands, and vascular smooth muscle cells as well as neurons are subject to this abnormally increased hormonal stimulation. The milk-induced change of the IGF-1-axis most likely contributes to the development of fetal macrosomia, induction of atopy, accelerated linear growth, atherosclerosis, carcinogenesis and neurodegenerative diseases. Observations of molecular biology are supported by epidemiologic data and unmask milk consumption as a promoter of chronic diseases of Western societies.